Department of Mechanical, Industrial, and Mechatronics Engineering
Toronto Metropolitan University
Common Mid-Term Test (W2025)
Materials Science I (MTL 200)

Test Duration: 90 mins

Name (Last-First):_____ **Section No:**
ID: **Instructor:**

Instructions:
1. **This is a closed-book test.**
2. **You may use an electronic calculator.**
3. **For multiple choice questions choose only the best answer.**
4. **Answer questions with a dark pen/pencil on the bubble sheet.**
5. **Back of pages can be used if extra pages are needed.**
6. **Students have to answer questions individually. No hints/ help will be provided.**

Questions 1-4, 6,7,11,14,17,19,20, 22-27, 29,30, 32-34,36, 38-41 (Each one mark)
Questions 5, 8-10,12,13,15,16, 18,21,28, 31,35,37 (Each two marks)
Questions 42-56 (Each three marks)

1. Metal **Na** (located in the first column of the periodic table) with an electronegativity value of 0.9 can make a strong bond with a non-metal **Cl** (located in the seventh column of the table) with an electronegativity value of 3.0. Which one of the following statements explains this ionic bonding:
(a) **Na** is an electronegative element and **Cl** is an electropositive element therefore, ionic bonding is formed between them.

(b) The high degree of electropositivity of **Na** and electronegativity of **Cl** creates such strong bonding.

2. Which of the following answers presents atomic bonding for MgO, CO_2, W, Ar, and NH_3, respectively.
(a) Ionic, van der Waals, Metallic, Covalent, and Hydrogen.

(b) Ionic, Covalent, Metallic, Covalent, and Hydrogen.

(c) Ionic , Covalent, Metallic, van der Waals, and Hydrogen.

(d) Ionic , Metallic, Covalent, van der Waals, and Hydrogen.

3. Quantum numbers are a set of four numbers that specify in sequence

(a) the size, shape, and spatial orientation of an electron's probability density, and spin orientation.

(b) the spin orientation, spatial orientation, size, and shape of an electron's probability density.

(c) the spatial orientation, spin orientation, shape and size of an electron's probability density.

(d) the size, shape, spin orientation, and spatial orientation of an electron's probability density.

4. Electronegativity value in the periodic table increases (1) and (2).

(a) (1) from right to left, (2) from top to bottom.

(b) (1) from right to left, (2) from bottom to top.

(c) (1) from left to right, (2) from top to bottom.

(d) (1) from left to right, (2) from bottom to top.

5. Which of the following electron configurations is for an inert gas?

(a) $1s^2\ s^2\ 2p^6\ 3s^2\ 3p^6$

(b) $1s^2\ 2s^2\ 2p^6\ 3s^2$

(c) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1$

(d) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^2\ 4s^2$

6. What type(s) of bonding would be expected for steel (an alloy composed of iron with a very small amount of carbon)?

(a) Ionic bonding

(b) Metallic bonding

(c) Covalent bonding with some van der Waals bonding

(d) van der Waals bonding

7. The type of interatomic bonding between a large agglomeration of carbon atoms is

(a) Ionic

(b) Covalent

(c) Metallic

(d) Permanent dipole

8. Aluminum has a modulus of elasticity that differs depending on the crystallographic orientation. This directional dependence is termed

(a) Anisotropy

(b) Polycrystallinity

(c) Polymorphism

(d) Allotropy

9. In FCC or BCC unit cells, interstitial atoms can occupy the following positions:
(a) (0,0,1), (1, ½, 0), (1, ½, ½)

(b) (0, ½,0), (1, ½, 1), (0, ½, 1)

(c) 0 ½ 0, 1 ½ 1, 0 ½ 1

(d) (½,½,1), (0,0, ½), (1, ½, 1)

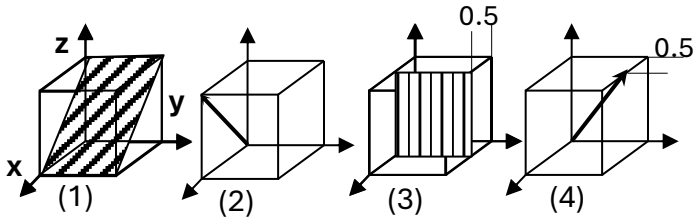
10. For Crystallographic directions and planes shown below which of the following answers is correct.

(a) 1: (101), 2: [101], 3: (200), 4: [122]

(b) 1: (110), 2: [110], 3: (0 ½ 0), 4: [112]

(c) 1: (101), 2: (101), 3: (½ 0 0), 4: [122]

(d) 1: [101], 2: (101), 3: (200), 4: [122]



11. Crystal structure represents how atoms or ions are arrayed in space. It is defined in terms of the unit cell (1) and the atom (2) within the unit cell.

(a) (1) direction, (2) sizes.

(b) (1) geometry, (2) positions.

(c) (1) plane, (2) positions.

(d) (1) geometry, (2) sizes.

12. The sufficient condition for diffraction to occur in the face-centered cubic crystal structure is:

(a) h + k + l must be even.

(b) h + k + l must be odd.

(c) h, k, and l must all be either odd or even.

(d) h, k, and l must all be neither odd nor even.

13. The closed-packed plane and stacking sequence of FCC crystal structures are

(a) (0001); ABCABCABC…
(b) (110); ABABAB…
(c) (111); ABCABCABC…
(d) (222); ABABABAB…

14. Anisotropy refers to materials where the properties are _____, while isotropy refers to materials where the properties are _____.

(a) Not directional; directional
(b) Directional, not directional
(c) Polycrystals, single crystals
(d) Non-dense and random, dense and ordered.

15. The close-packed planes and directions for an FCC structure are

(a) (110), [111]

(b) (111), [110]

(c) [111], (111)

(d) [110], (112)

16. For BCC Alpha Iron, with $(h^2+k^2+l^2) = 12$, the miller indices are

(a) (222)

(b) (310)

(c) (311)

(d) (312)

17. In steel containing 99.5% iron and 0.5% carbon,

(a) Iron atoms represent the solvent element and carbon atoms represent the solute element.

(b) Iron atoms represent the solute element and carbon atoms represent the solvent element.

18. Burger's vector presents:

(a) The magnitude of material deformation.

(b) Both the magnitude and the direction of material deformation.

(c) The direction of dislocation motion.

(d) The lattice distance in a crystal structure.

19. Grain boundaries are (1) chemically reactive than the grains themselves because of the (2) energy state of grain boundaries.

(a) (1) more, (2) lower.

(b) (1) more, (2) higher.

(c) (1) less, (2) lower.

(d) (1) less, (2) higher.

20. Twinning occurs on (1) crystallographic plane and in a (2) direction.

(a) (1) a definite, (2) general.

(b) (1) a definite, (2) specific.

(c) (1) an indefinite, (2) general.

(d) (1) an indefinite, (2) specific.

21. The conditions for a substitutional solid solution are

(a) Δr (atomic radius) < 15%, proximity in the periodic table (electronegativities), same crystal structure, and valency.

(b) Δr (atomic radius) > 15%, Burgers vector, density, and bonding strength.

(c) Δr (atomic radius) < 15%, solubility, valency, and number of slip systems.

(d) Δr (atomic radius) < 15%, grain size, same crystal structure, and valency.

22. Exhibition of mirror lattice symmetry, where atoms on one side are in mirror-image positions to those atoms on the other side, it is a

(a) Phase boundary

(b) Twist boundary

(c) Tilt boundary

(d) Twin boundary

23. As the number of grain boundaries in a material increases, what is the effect on yield strength?

(a) Increases

(b) Decreases

(c) Stays the same

24. If the grains in polycrystalline materials have preferred crystallographic orientations, the material has a

(a) stacking fault

(b) texture

(c) twin

(d) slip

25. A system is at equilibrium if its free energy is

(a) Maximum

(b) Minimum

(c) Steady

(d) Varying

26. Diffusion is the atomic motion in a predictable fashion and leads
(a) to create a steady-state condition.

(b) to eliminate concentration differences and to produce a uniform composition.

(c) to transfer the amount of energy from one atom to another atom.

(d) to create an irregular atomic structure.

27. The energy required to move an atom from a normal atomic position to another normal site is greater than that of moving an interstitial atom. This may verify that
(a) Substitutional atomic diffusion is less stable than interstitial atomic diffusion.

(b) Substitutional atomic diffusion is more stable than interstitial atomic diffusion.

(c) The magnitude of energy is independent of the stability of atoms against motion.

28. Interstitial diffusion takes place (1) more rapidly than vacancy diffusion because the interstitial atoms are smaller and (2) mobile. Also, there are (3) vacant interstitial sites than vacancies.

(a) (1) less, (2) more, (3) more.

(b) (1) more, (2) less, (3) more.

(c) (1) more, (2) more, (3) less.

(d) (1) more, (2) more, (3) more.

29. Carburizing is a process by which the surface (1) concentration of a ferrous alloy is (2) by diffusion from the surrounding environment at high temperatures.

(a) (1) carbon, (2) increased.

(b) (1) carbon, (2) decreased.

(c) (1) oxygen, (2) increased.

(d) (1) oxygen, (2) decreased.

30. Relative to vacancy diffusion, interstitial diffusion is

(a) less rapid
(b) more rapid
(c) dormant
(d) ineffective

31. The negative sign in Fick's first law indicates that diffusion occurs

(a) from a low to high concentration
(b) is non-steady state
(c) from a high to a low concentration
(d) not occurring

32. This type of diffusion is not subject to local compositional changes.

(a) Interdiffusion

(b) Self-diffusion

(c) Impurity diffusion

(d) Interstitial diffusion

33. In iron-carbon alloys, self-diffusion of iron occurs by

(a) Vacancy mechanism

(b) Interstitial mechanism

(c) Steady-state mechanism

(d) Non-steady-state mechanism

34. A system is at equilibrium if

(a) All phases of the system consist of the same crystal structure.

(b) All phases of the system consist of the same concentrations at a constant temperature.

(c) The system includes the minimum randomness or disorder of the atoms.

(d) The system possesses the highest energy and thermal energy.

35. Binary phase diagram (Ni-45% Cu) is shown.

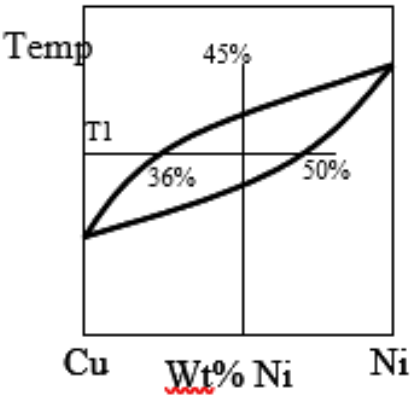
The tie line at T1 Temperature represents:

(a) The solid phase is first nucleated when it contains 36% Ni.

(b) Liquid contains 36% Ni and solid contains 50% Ni.

(c) Both (a) and (b)

(d) None of the above



36. In the solidification process of an isomorphous alloy during equilibrium cooling, the compositions of liquid and solid phases in the two-phase region will follow

(a) solidus and liquidus lines, respectively.

(b) liquidus and solidus lines, respectively.

(c) liquidus and solvus lines, respectively.

(d) solvus and liquidus lines, respectively.

37. Eutectic structure is a (1) microstructure consisting of alternating layers (or lamellae) of the phases that form (2) during the solidification of an alloy having the eutectic composition.

(a) (1) two-phase, (2) consecutively.

(b) (1) two-phase, (2) simultaneously.

(c) (1) one-phase, (2) consecutively.

(d) (1) one-phase, (2) simultaneously.

38. Upon heating, one solid phase transforms into a liquid phase and another solid phase. This is a _____ reaction.

(a) eutectic
(b) eutectoid
(c) peritectic
(d) monotectic

39. An alloy composition between that of the right-hand-side end of the tie line defining the eutectic reaction and the eutectic composition is known as

(a) hypoeutectic alloy
(b) eutectic alloy
(c) hypereutecoid alloy
(d) hypereutectic alloy

40. Phase transformations may involve

(a) Transition from one phase to another

(b) Appearance of a phase

(c) Disappearance of a phase

(d) All of the above

41. Upon heating, one solid phase transforms into a liquid phase and another solid phase. This is a _____ reaction.

(a) eutectic reaction

(b) eutectoid reaction

(c) peritectic reaction

(d) monotectic

(e) diagonal

42. Identify the element having an inter-planar spacing of d_{110}=0.2237 nm, and (a/R)=2.828 (use: $d = \frac{a}{\sqrt{h^2+k^2+l^2}}$)

(a) Nickel with an atomic radius of 0.125 nm.

(b) Cadmium with an atomic radius of 0.197 nm.

(c) Manganese with an atomic radius of 0.112 nm.

(d) Tungsten with an atomic radius of 0.137 nm.

43. The radius of an iridium atom, given that Ir has an FCC crystal structure, a density of 22.4 g/cm^3, and an atomic weight of 192.2 g/mol.

$$\rho = \frac{nA_{\mathrm{Ir}}}{V_C N_{\mathrm{A}}}$$

(a) 0.136 nm

(b) 1.45 nm

(c) 2.45 nm

(d) 0.136 mm

44. The diffraction angle of an FCC metal (λ = 0.1542 nm, r = 0.1387 nm) from the (113) set of planes is ($d = \frac{a}{\sqrt{h^2+k^2+l^2}}, 2d\sin\theta = n\lambda$)

(a) 81.4°
(b) 39°
(c) 40.7°
(d) 78°

45. For Platinum (d = 0.1138 nm) with a wavelength of 0.1542 nm, the diffraction angles will be, in degrees ($n\lambda = 2dsin\theta$)

(a) 60.12

(b) 50.48

(c) 74.80

(d) 85.30

46. The magnitude of Burger's vector along direction [110] for Copper with FCC structure and lattice distance of 0.36151×10^{-8}cm is calculated to be ($|b| = \frac{a}{2}\sqrt{h^2 + k^2 + l^2}$):

(a) 6.55×10^{-8} cm

(b) 0.55×10^{-8} cm

(c) 5.55×10^{-8} cm

(d) 2.55×10^{-8} cm

47. The ASTM grain size number for 25 grains per square inch is measured at a magnification of 600X

$$n_M\left(\frac{M}{100}\right)^2 = 2^{G-1}$$

(a) 2
(b) 5
(c) 8

(d) 10.8

48. Calculate the number of vacancies per cubic meter at 1000°C=1273K for a metal that has an energy for vacancy formation of 1.22 eV/atom, a density of 6.25 g/cm^3, and an atomic weight of 37.4 g/mol. ($N_v = N\exp\left(-\frac{Q_v}{RT}\right)$; $N = \frac{N_A\rho}{A}$)

(a) 1.49 × 1018 m^{-3}
(b) 7.18 × 1022 m^{-3}
(c) 1.49 × 1024 m^{-3}
(d) 2.57 × 1024 m^{-3}

49. The number of atoms diffusing through a membrane with an area of 4 cm^2 is 6.16×10^{13} atoms/sec (based on the definition of flux =M/A.t). The concentration gradient ($\Delta c/\Delta x$) for diffusing atoms with $D=9\times10^{-12}$ cm^2/sec becomes: (Note: $J=-D(\Delta c/\Delta x)$)

(a) -1.71×10^{24}

(b) -1.71×10^{23}

(c) -1.71×10^{22}

(d) -1.71×10^{21}

50. A sheet of steel 1.5 mm thick has nitrogen atmospheres on both sides at 1200°C and is permitted to achieve a steady-state diffusion condition. The diffusion coefficient for nitrogen in steel at this temperature is 6×10^{-11} m^2/s, and the diffusion flux is found to be 1.2×10-7 kg/m^2.sec. Also, it is known that the concentration of nitrogen in the steel at the high-pressure surface is 4 kg/m^3. How far into the sheet from this high-pressure side will the concentration be 2.0 kg/m^3? Assume a linear concentration profile.

$$J = -D\frac{C_A - C_B}{x_A - x_B}$$

(a) 100 mm

(b) 50 mm

(c) 5 mm

(d) 1 mm

51. Determine the diffusion coefficient for copper in aluminum at 600°C. The pre-exponential and activation energy values for this system are 6.5×10^{-5} m^2/sec and 136,000 J/mol, respectively. ($D = D_0 \exp\left(-\frac{Q_d}{RT}\right)$)

(a) 5.7×10^{-2} m^2/s
(b) 4.7×10^{-13} m^2/s
(c) 9.4×10^{-17} m^2/s
(d) 3.9×10^{-2} m^2/s

52. The diffusion coefficients for copper in aluminum at 500 and 600 C are 4.8×10^{-14} and 5.3×10^{-13} m^2/s, respectively. Determine the approximate time at 500 C that will produce the same diffusion result at 600°C (x is constant at both temperatures) as a 10hr heat treatment. ($\frac{x^2}{D.t} = Constant$)

(a) 35 hrs

(b) 40 hrs

(c) 129.5 hrs

(d) 110.4 hrs

53. Given $D=3\times10^{-11}$ m^2/s and $\Delta C = 0.4$ kg/m^3 measured between carbon positions at 5 and 10 mm, the diffusion flux (kg/m^2s) will be?

$$J = -D\frac{dC}{dx} = -D\frac{C_A - C_B}{x_A - x_B}$$

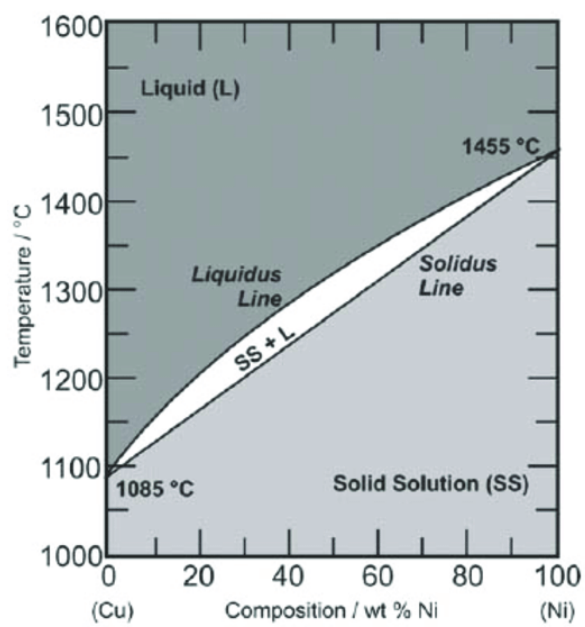
(a) 2.1×10^{-9}

(b) 2.2×10^{-9}

(c) 2.4×10^{-9}

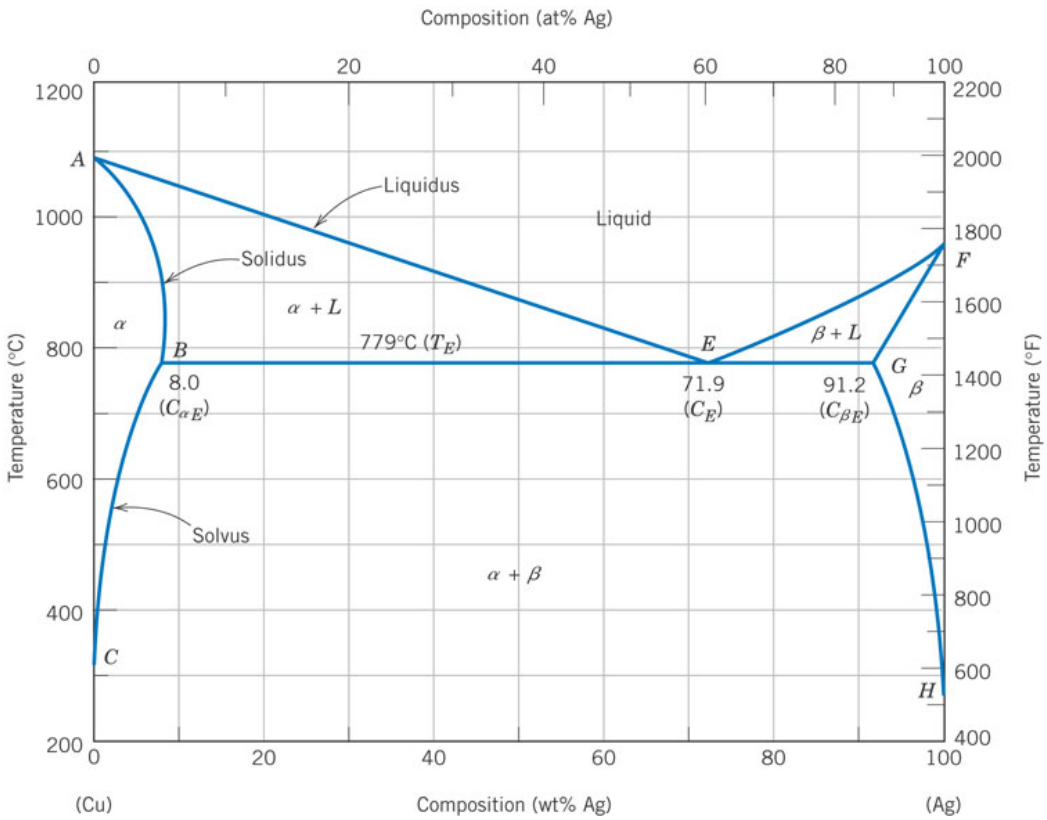
(d) 2.8×10^{-9}

54. Which statement is true for a Copper–Nickel alloy isomorphous phase diagram, at equilibrium?



(a) Consists of a liquid phase of composition 20 wt% Ni–80 wt% Cu and also has a solid solution(SS) phase of composition 40 wt% Ni–60 wt% Cu
(b) At 1300°C and 45wt% Cu-55wt% Ni, both solid solution and liquid phase exist.
(c) Consists of a liquid phase of composition 10 wt% Ni–90 wt% Cu and also has a solid solution(SS) phase of composition 40 wt% Ni–60 wt% Cu
(d) At 1200°C and 45wt% Cu-55wt% Ni only liquid phase exists

The Copper-Silver binary system is given below. Using the binary system, <u>answer questions 55 and 56.</u>



55. For an alloy with a composition of 25 wt% Ag-75 wt% Cu, the wt.% of primary α that forms under equilibrium cooling at eutectic temperature is:

(a) 73.4 wt%.
(b) 88.6 wt%.
(c) 37.4 wt%.
(d) 47.3 wt%.

56. For this alloy with a composition of 25 wt% Ag-75 wt% Cu at 775°C (1425°F), the mass fractions of α and β phases are:

(a) 0.796 and 0.204 respectively
(b) 0.734 and 0.266 respectively
(c) 0.912 and 0.8 respectively
(d) 91.2 and 8.0 respectively